Virtual Work or Unit Load Method

The method of virtual work is the most fundamental, direct and versatile tool for computing deflection of structures. The method can be used to evaluate deflections resulting from loads of any type, temperature changes, error in fabrications and movement of supports. The method is based on Bernoulli's principle of virtual displacements which is stated as follows:

> *If a rigid body under the action of any set of forces in equilibrium be given any small virtual displacement, the total work of the force system (virtual work) is equal zero.*

The corresponding theorem for deformable bodies known as the principle of virtual work is stated as follows:

> *If a deformable body, in equilibrium under the action of a system of loads, is subjected to a small virtual displacement, the external virtual work done by the external forces is equal to the internal virtual work of the stress caused by the given system of loads.*

The equality is written as follows:

External Virtual Work = Internal Virtual Work

Work of External Forces

The work done by an external force P as it rides along at its full value during such a virtual displacement Δ is called virtual work, which is expressed as:

External virtual work: $\mathbf{W_e = P\Delta}$

In the same manner, the virtual work done by a moment M during a virtual angular rotation θ is expressed as:

External virtual work: $\mathbf{W_e = M\theta}$

In general, the total virtual work done by a force system will equal the sum of the product of each load by the corresponding virtual displacement and generalized as:

External virtual work: $\mathbf{W_e = \Sigma P\Delta + \Sigma M\theta}$

Internal Virtual Work

The internal virtual work for any element of the deformed structure is expressed as the product of the average stress on the element and the corresponding virtual deformation of the element in the direction of the stress. Expressions will be developed for the internal virtual work of axial and flexural strains, which are then equated to their respective external virtual works to solve for the resulting deflection component.

Displacement due to Axial Strains: (for Truss Type structures)

Consider a prismatic bar of length L, cross sectional area A and made of a material with modulus of elasticity E, subjected to an axial force N, the resulting axial deformation is:

$$\Delta = \frac{NL}{EA}$$

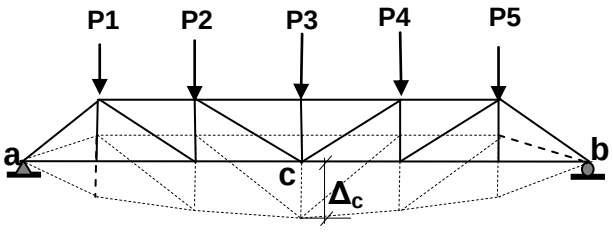
If this deformation is considered to be a virtual displacement of the bar, the internal virtual work done by a bar force resulting from the action of another fictitious unit axial force ***"n"*** as it rides along at its full value is;

$$W_i = n\frac{NL}{EA}$$

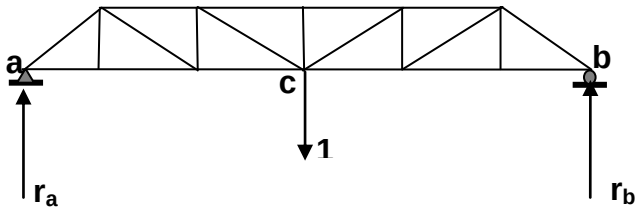
For example in computing truss deflection, two different states are considered:

State I – The truss acted upon by the real loads is considered (fig. a).

State II – The truss acted upon by a unit fictitious load at a point in the direction of required deflection (fig. b).



(a) State I



(a) State II

If for example the vertical displacement at point C of the truss is required; assume that a unit load in the direction of the required displacement is applied to the truss which is in equilibrium (state II), then the actual displacement of the truss (state I) can be considered to be virtual displacement to the effect of unit force of state II.

- The external virtual work done due to the effect of fictitious unit load through the displacement will be: $\mathbf{W_e = \Sigma P\Delta = r_a * 0 + 1*\Delta + r_b * 0 = 1*\Delta}$
- The internal virtual work in each member of the truss will be: $n\frac{NL}{EA}$. Thus the total internal virtual work for the entire truss will be:

$$W_i = \sum\left(n\frac{NL}{EA}\right)$$

By the principle of virtual work:

$$W_e = W_i$$

$$1\times\Delta = \sum\left(n\frac{NL}{EA}\right)$$

Where:

n = axial force in members of the truss due to a unit fictitious load applied at the point where the deflection is sought along the desired direction.

N = axial force in members of the truss due to the actual loads

E, A, L = modulus of elasticity, cross-sectional area and length of members of the truss respectively.

* Note that the member forces n due to a fictitious unit load is a deflection coefficient and when multiplied by the change of length (due to any cause) of the corresponding member will give the effect of the change in length of the members on the required deflection component, thus the above expression can be generalized as follows:

$$\mathbf{1*\Delta = \Sigma\ n\,\delta L}$$

where: δL = change in length

= NL/EA, due to applied forces (i)

= $\alpha(\Delta T)L$, due to temperature change (ii)

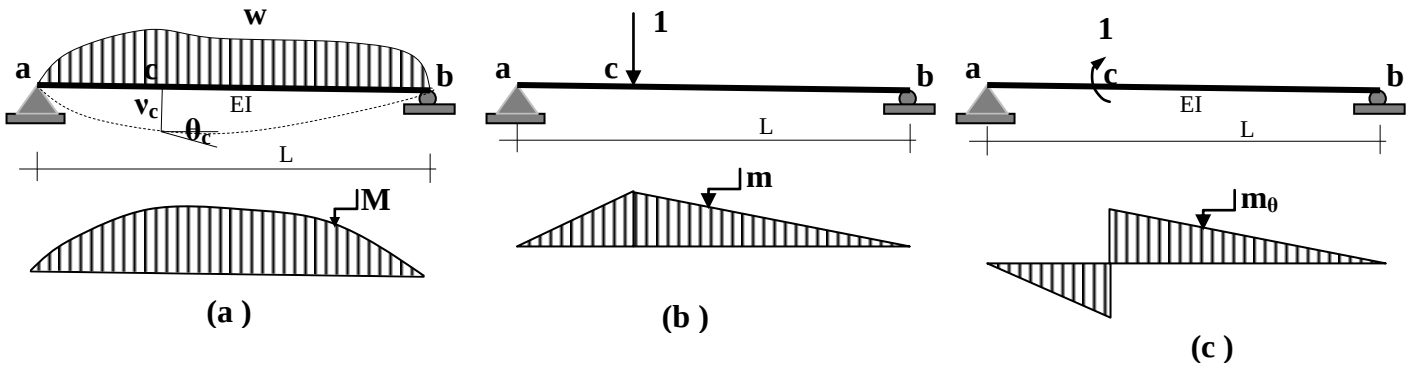
= Specified value; such as errors in fabrication (iii)

n = deflection coefficient

Displacement Due to Flexural Strain

The lateral displacement and rotation at any point on a beam due to flexural strain/bending moments can be obtained using the principle of virtual work by applying a unit fictitious force and couple respectively at the point and equating the external and internal virtual work.

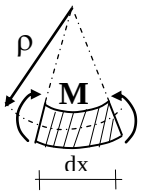
Consider a simply supported beam subjected to transverse loads as shown below.



To compute the vertical displacement at a point (point c), apply a unit vertical force at the point as shown in fig (b) above.

- The curvature of a segment of the beam *dx* subjected to a bending moment M due to the real load is given by:

$$\frac{1}{\rho} = \frac{d\nu^2}{dx^2} = \frac{d}{dx}\left(\frac{d\nu}{dx}\right) = \frac{d\theta}{dx} = \frac{M}{EI} \quad \Rightarrow d\theta = \frac{M}{EI}dx$$



- The internal bending moment at section *x*, caused by the unit fictitious load designated by *m*, performs an internal virtual work in segment *dx* which is equal to:

$$dW_i = m d\theta = m\frac{M}{EI}dx$$

The total internal virtual work resulting from flexural strain in the total length L of the beam is:

$$W_i = \int_L dW_i = \int_L m d\theta = \int_L m\frac{M}{EI}dx$$

The external virtual work of the unit load due to the displacement Δ , which is virtual to the unit load is

$$W_e = 1^* \Delta$$

- From the principle of virtual work $W_e = W_i$

$$1 \times \Delta = \int_L m \frac{M}{EI} dx$$

To compute the rotation at a point (point c), apply a unit fictitious couple at the point as shown in fig (c) above and using the same procedure as for the lateral deflections. It can be shown that:

$$1 \times \theta = \int_L m_\theta \frac{M}{EI} dx$$

Where:

m = bending moment due to a unit fictitious lateral load

m_θ = bending moment due to a unit fictitious couple

M = bending moment due to real loads

EI = flexural stiffness of the beam

Graphical Multiplication Method

The above integral contains the product of two ordinates of the m and M curves obviously the m curves are straight line. Hence the integration of the product can be carried out in such a way that:

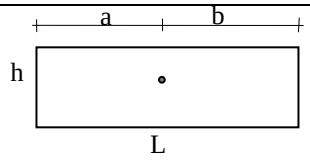
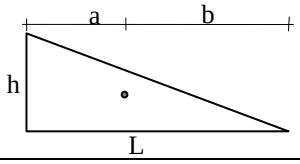
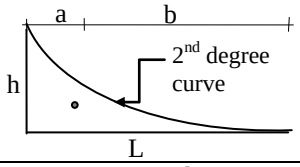
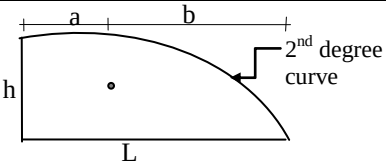
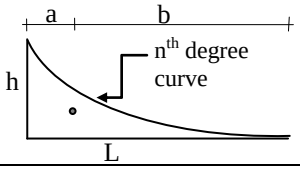
$$\int_L m \frac{M}{EI} dx = A_m m_c$$

Where:

A_m = the area bounded by the M/EI diagram b/n the given limit.

m_c = the ordinate of the m diagram at a point corresponding to the centroid of the M/EI diagram b/n the given limits.

The area and position of centroids for most commonly encountered shapes are found to be as follows:

Shape of Diagram	Area	Position of Centroid	
		a	b
	A = hL	L/2	L/2
	A = hL/2	L/3	2L/3
	A = hL/3	L/4	3L/4
	A = 2hL/3	3L/8	5L/4
	A = hL/(n+1)	L/(n+2)	(n+1)L/(n+2

In general, the internal virtual work in a frame work consisting of members which are subjected to both axial force and bending moments can be considered as the sum of the work done due to the axial strain and flexural strains. Hence the principle of virtual work can be generalized as:

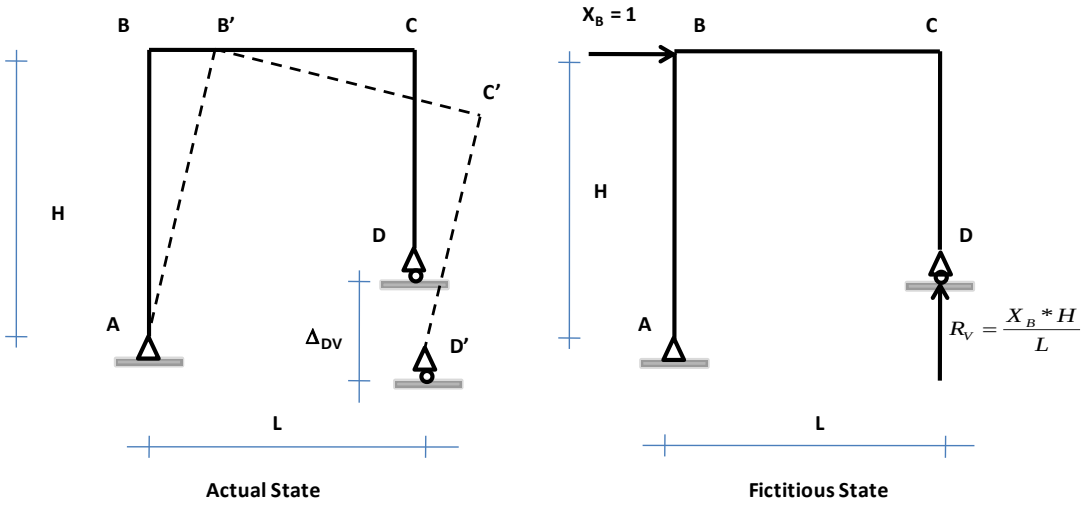
$$1 \times \Delta = \sum \left(n \frac{NL}{EA} \right) + \int_L \mathrm{m} \frac{\mathrm{M}}{\mathrm{EI}} \mathrm{dx}$$

**** Note that in most of the cases for continuous beams and rigid frames, the effect of axial forces for deflections are very small as compared to deflection due to bending moment and thus are usually neglected.***

Displacement due to Support Movement

It is evident that no internal stresses will result from movements of supports of statically determinate structures. Thus, the internal virtual work in such cases is equal to zero. Accordingly, by the principle of virtual work, the external virtual work is seen to be zero.

For example consider the following frame, where the horizontal displacement at B due to a vertical settlement of Support D is required.



(Fig. a) (Fig. b)

Assume a unit horizontal force $X_B = 1$ is applied at point B (fig. b)

- The actual state of the structure carries no load at all implying that the internal virtual work done by the stresses due to a unit fictitious force is zero.

- By the principle of virtual work, the external virtual work is zero.

$$X_B * \Delta_{BH} - R_{DV} * \Delta_{DV} = 0$$

And since $X_B = 1$ and $R_D = H/L$

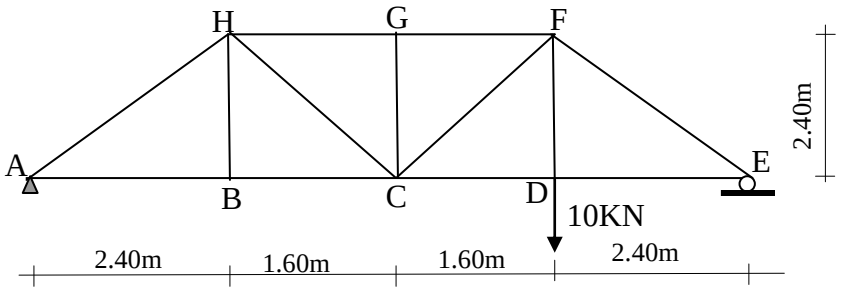
$$1* \Delta_{BH} - (H/L) * \Delta_{DV} = 0 \Rightarrow \mathbf{\Delta_{BH} = (H/L) * \Delta_{DV}}$$

In general, the displacements indicated in a statically determinate structure due to the movement of its supports can be determined as per the following procedures:

a. *Choose a fictitious state of the structure for which the support under consideration remains fixed.*
b. *Apply a unit load to the structure in the fictitious state in the same direction corresponding to the required displacement.*
c. *Determine the reaction produced by the applied unit load at the support in the same direction as the support movement.*
d. *Find the total virtual work done by the loads and reactions of the fictitious state and equate to zero.*
e. *Determine the displacement required by solving the virtual work equation.*

Examples

1. Find the vertical deflection of joint B for the truss due to the applied load using the virtual work method. All external members have cross-sectional area of 6cm2 and all internal members have cross-sectional area of 4cm^2. Use E =210 G Pa for all members.



Soln: - ***analyze the truss for the given loads & determine the bar forces***

- ***apply a unit vertical load at joint B and compute the forces in the member due to the unit load independently.***
- ***Apply the principle of virtual work.***
- ***The result is presented more conveniently in tabular form as follows.***

Member	L (m)	A (cm^2)	N (KN)	n	$n\frac{NL}{A}$
AB	2.4	6	+3.00	+0.70	+0.84
BC	1.6	6	+3.00	+0.70	+0.56
CD	1.6	6	+7.00	+0.30	+0.56
DE	2.4	6	+7.00	+0.30	+0.84
EF	3.394	6	-9.90	-0.424	+2.374
FG	1.6	6	-5.00	-0.50	+0.667
GH	1.6	6	-5.00	-0.50	+0.667
HA	3.394	6	-4.24	-0.99	+0.667
HB	2.40	4	0.00	+1.00	0
HC	2.884	4	+3.61	-0.361	-0.940
GC	2.40	4	0.00	0.00	0
CF	2.884	4	-3.61	+0.361	-0.940
FD	2.400	4	+10.00	0.00	0
					Σ= 7.002

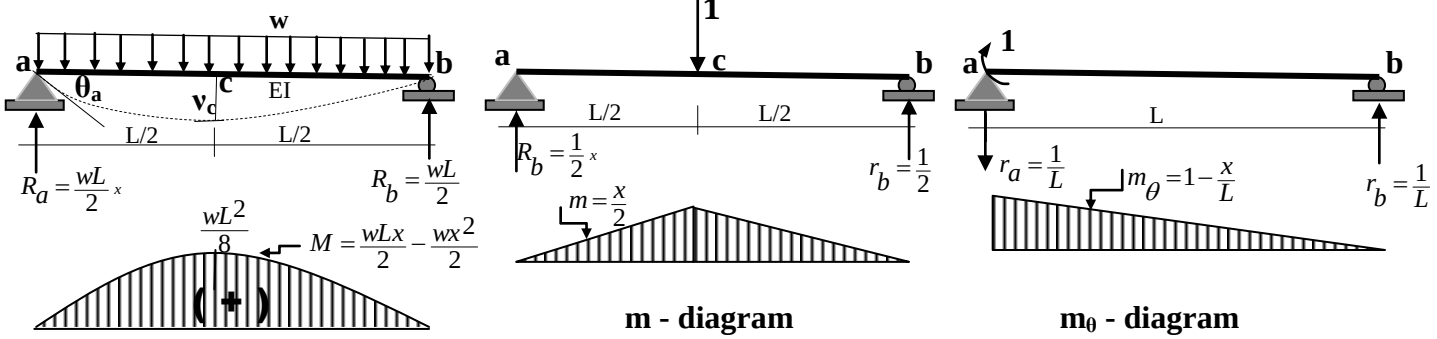
$$1\times\Delta_B=\Sigma\left(n\frac{NL}{EA}\right)$$

$$\Delta_B=\frac{7.002}{210}*10$$

Δ_B = 0.333 mm , downward

2. Using virtual work method, determine the vertical displacement and rotation at specified points for the following simple cases of beams

(a) ν_c = ?
θ_a = ?



M - diagram **m - diagram** **m_θ - diagram**

- . Displacement at c:

Internal Virtual work:

$$W_i = 2\int_0^{L/2} m\frac{M}{EI}dx = 2\int_0^{L/2} \frac{x}{2EI}\left(\frac{wLx}{2}-\frac{wx^2}{2}\right)dx = \frac{1}{2EI}\left[\frac{wLx^3}{3}-\frac{wx^4}{4}\right]_0^{L/2} = \frac{5wL^4}{384EI}$$

External Virtual work: $W_e = 1X\,\nu_C = \nu_C$

By the principle of virtual work, $W_e = W_i$ $\longrightarrow$ $\nu_c = = \frac{5wL^4}{384EI}$, *downward*

- . Rotation at a:

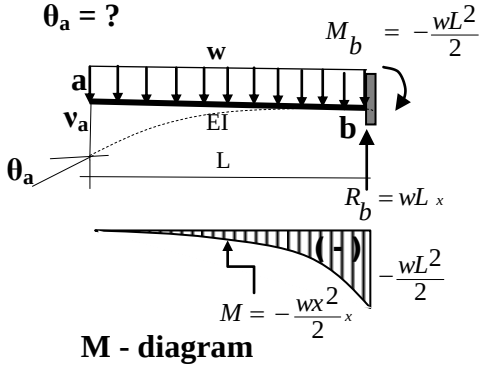
Internal Virtual work:

$$W_i = \int_0^{L} m_\theta \frac{M}{EI}dx = \frac{1}{EI}\int_0^{L}\left(1-\frac{x}{L}\right)\left(\frac{wLx}{2}-\frac{wx^2}{2}\right)dx = \frac{w}{2EI}\left[\frac{Lx^2}{2}-\frac{x^3}{3}\right]_0^{L} = \frac{wL^3}{24EI}$$

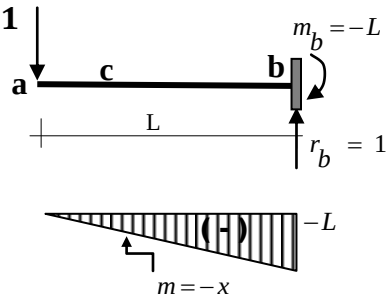
External Virtual work: $W_i = 1X\,\theta a = \theta a$

By the principle of virtual work, $W_e = W_i$ $\longrightarrow$ $\theta a = \frac{wL^3}{24EI}$, *clockwwise*

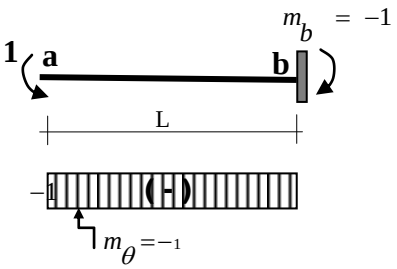
(b) ν_a **= ?**
θ_a **= ?**



M - diagram



m - diagram



m_θ - diagram

- . Displacement at a:

Internal Virtual work: $W_i = \int_0^L m\frac{M}{EI}dx = \frac{1}{EI}\int_0^L (-x)\left(-\frac{wx^2}{2}\right)dx = \frac{1}{EI}\left[\frac{wx^4}{8}\right]_0^L = \frac{wL^4}{8EI}$

External Virtual work: $W_i = 1X\,\nu_a = \nu_a$

By the principle of virtual work, $W_e = W_i$ $\longrightarrow$ $\nu_a = \frac{wL^4}{8EI}$, *downward*

- . Rotation at a:

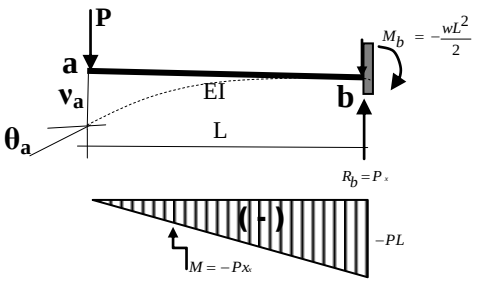
Internal Virtual work:

$W_i = \int_0^L m_\theta\frac{M}{EI}dx = \frac{1}{EI}\int_0^L (-1)\left(-\frac{wx^2}{2}\right)dx = \frac{1}{EI}\left[\frac{wx^3}{6}\right]_0^L = \frac{wL^3}{6EI}$, *cownter clock wise*

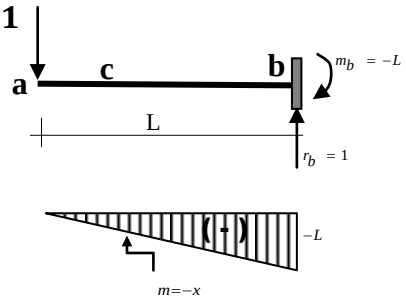
External Virtual work: $W_e = 1X\,\theta_C = \theta_C$

By the principle of virtual work, $W_e = W_i$ $\longrightarrow$ $\theta_c = \frac{wL^3}{6EI}$, *cownter clock wise*

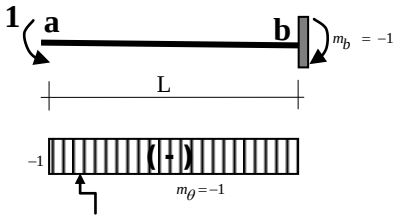
(c) $\mathbf{v_a = ?}$
$\mathbf{\theta_a = ?}$



M - diagram



m - diagram



$\mathbf{m_\theta}$ - diagram

- . Displacement at a:

Internal Virtual work: $W_i = \int_0^L m\frac{M}{EI}dx = \frac{1}{EI}\int_0^L (-x)(-Px)dx = \frac{1}{EI}\left[\frac{Px^2}{3}\right]_0^L = \frac{PL^3}{3EI}$

External Virtual work: $W_i = 1X\,v_a = v_a$

By the principle of virtual work, $W_e = W_i$ $\longrightarrow$ $v_a = \frac{PL^3}{3EI}$, *downward*

- . Rotation at a:

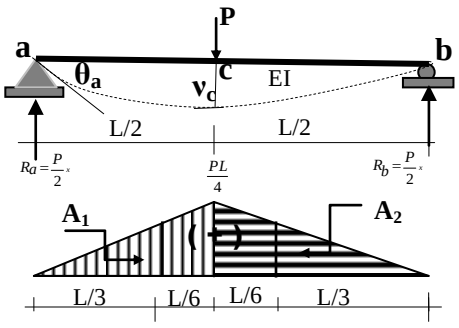
Internal Virtual work:

$W_i = \int_0^L m_\theta \frac{M}{EI}dx = \frac{1}{EI}\int_0^L (-1)(-Px)dx = \frac{1}{EI}\left[\frac{Px^2}{2}\right]_0^L = \frac{PL^2}{2EI}$, *cownter clock wise*

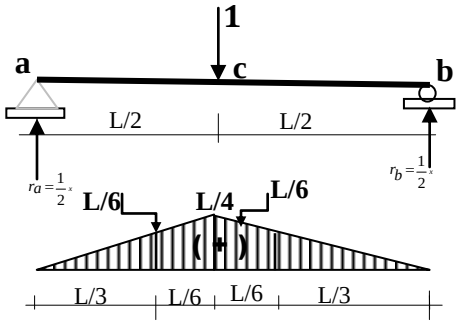
External Virtual work: $W_e = 1X\,\theta_C = \theta_C$

By the principle of virtual work, $W_e = W_i$ $\longrightarrow$ $\theta_c = \frac{PL^2}{2EI}$, *cownter clock wise*

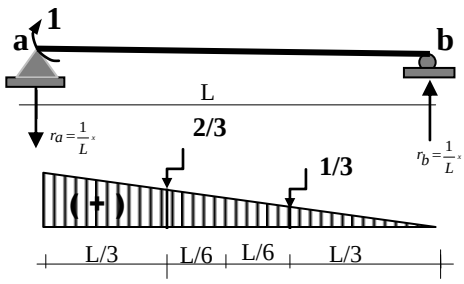
(d) v_c = ?
θ_a = ?



M - diagram



m - diagram



m_θ - diagram

- . Displacement at c:

Internal Virtual work: $W_i = \int_0^L m\frac{M}{EI}dx = \sum m_C A_M = \frac{2}{EI}\left[\frac{1}{2}*\frac{PL}{4}*\frac{L}{2}*\frac{L}{6}\right] = \frac{PL^3}{48EI}$

External Virtual work: $We = 1X\,v_C = v_C$

By the principle of virtual work, $W_e = W_i$ $\longrightarrow$ $v_c = = \frac{PL^3}{48EI}$, *downward*

- . Rotation at a:

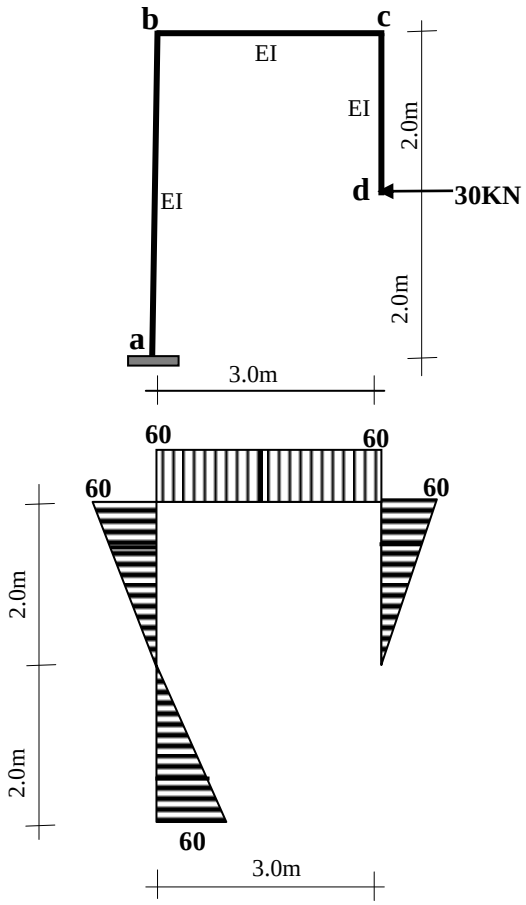
Internal Virtual work:

$$W_i = \int_0^L m_\theta \frac{M}{EI}dx = \sum m_{\theta,c} A_M = \left(\frac{2}{3}*\frac{PL/4*L/6}{2EI}\right) + \left(\frac{1}{3}*\frac{PL/4*L/6}{2EI}\right) = \frac{PL^2}{16EI}$$

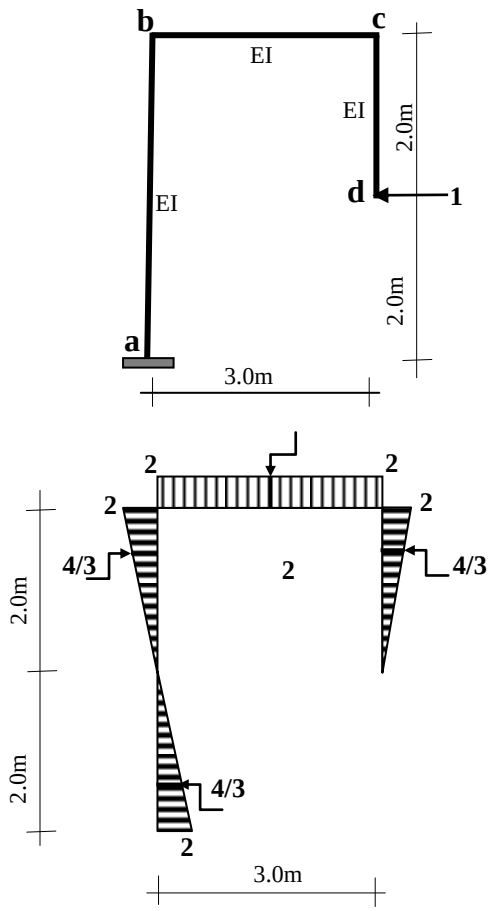
External Virtual work: $W_e = 1X\,\theta a = \theta a$

By the principle of virtual work, $W_e = W_i$ $\longrightarrow$ $\theta_a = \frac{PL^2}{16EI}$, *clockwwise*

3. Using virtual work method, determine the horizontal displacement of joint d for the following frame



M – diagram on tension side (KNM)



m – diagram on tension

- . Horizontal Displacement at d:

Internal Virtual work:

$$W_i = \int_0^L m\frac{M}{EI}dx = \sum m_c A_M = \frac{1}{EI}\left[3\left(\frac{1}{2}*60*2*\frac{4}{3}\right) + 60*3*2\right] = \frac{600}{EI}$$

External Virtual work: $We = 1X\,\Delta_{dh} = \Delta_{dh}$

By the principle of virtual work, $W_e = W_i$ $\longrightarrow$ $\Delta_{dh} = \frac{600}{EI}$, *to the left*

Exercise

1. Determine the vertical deflection of joint **c** for the truss loaded as shown in Fig. 1. The cross-sectional area of all external members and all internal members are $8cm^2$ and $6cm^2$ respectively. Use E = 210GPa for all members
2. Calculate the horizontal component of deflection of joint **g** for the truss loaded as shown in Fig.2. All members are made of the same material with modulus of Elasticity E = 210GPa and have cross-sectional areas indicated in brackets along with each member
3. Determine the vertical deflection at points **b** and **e** for beam loaded as shown in Fig. 3
4. Find the horizontal displacement at **d** and rotation at **b** for the frame loaded as shown in Fig. 4

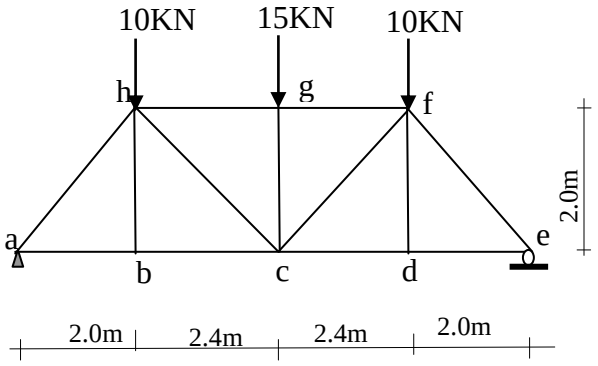


Fig. 1

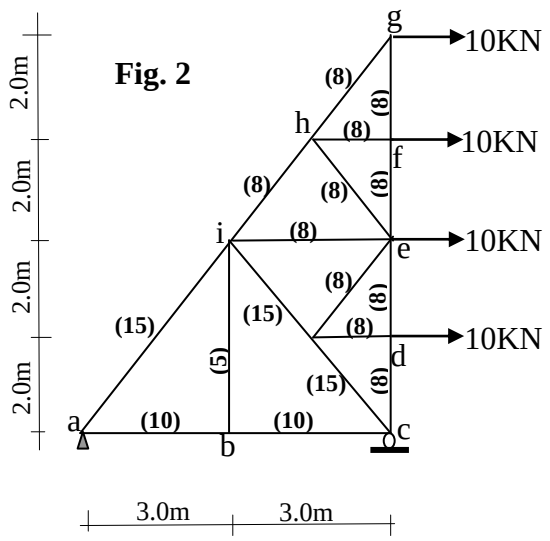


Fig. 2

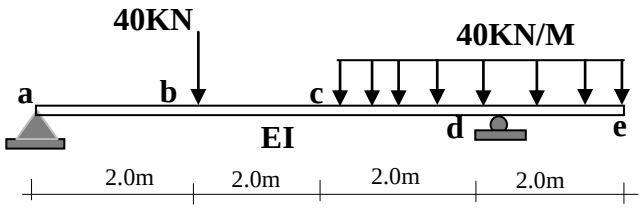


Fig. 3

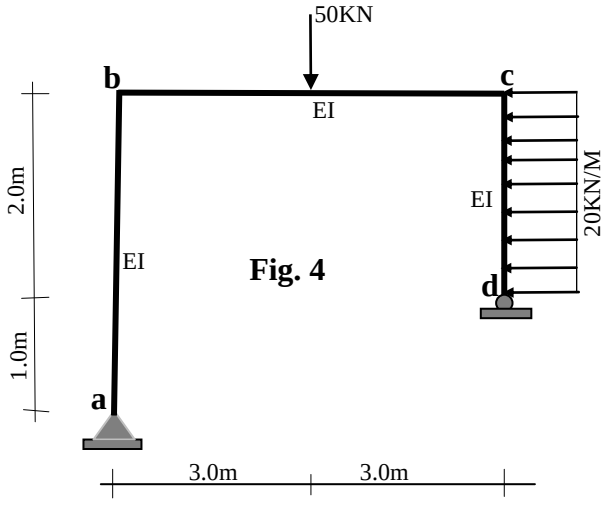


Fig. 4